

1. In how many ways can a committee of 5 be formed from a group of 11 people consisting of 4 teachers and 7 students if
 - (i) the committee must include exactly 2 teachers?
 - (ii) the committee must include at least 3 teachers?
 - (iii) a particular teacher and a particular student cannot be both in the committee?
2. There are 7 boys and 3 girls in a gathering. In how many ways can they be arranged in a row so that
 - (i) the 3 girls form a single block?
 - (ii) the two end-positions are occupied by boys and no girls are adjacent?
3. m boys and n girls are to be arranged in a row. Find the number of ways this can be done in each of the following cases:
 - (i) no boys are adjacent ($m < n + 1$);
 - (ii) the n girls form a single block;
 - (iii) a particular boy and a particular girl must be adjacent.
4. There are 12 students in a party. Five of them are girls. In how many ways can these 12 students be arranged in a row if
 - (i) the 5 girls must be together?
 - (ii) no 2 girls are adjacent?
 - (iii) between two particular boys A and B , there are no boys but exactly 3 girls?
5. Three girls A , B and C , and nine boys are to be lined up in a row. In how many ways can this be done if B must lie between A and C , and A , B must be separated by exactly 4 boys?
6. In how many ways can 5 boys and 3 girls be seated around a table if
 - (i) boy B_1 and girl G_1 are not adjacent?
 - (ii) no girls are adjacent?
7. Let S be the set of natural numbers whose digits are chosen from $\{1, 3, 5, 7\}$ such that no digits are repeated. Find number of elements of S and sum of the elements of S .
8. Using the numbers 1, 2, 3, 4, 5 we can form $5! (= 120)$ 5-digit numbers in which the 5 digits are all distinct. If these numbers are listed in increasing order: 12345, 12354, 12435, ... Find the position of the number 35421 and the 100 th number in the list.
9. Show that $(4n)!$ is a multiple of $2^{3n}3^n$, for each natural number n .
10. Prove that $\binom{n}{r} = \binom{n-1}{r-1} + \binom{n-1}{r}$.

11. Given positive integers r and n with $r \leq n$, let $\left[\begin{smallmatrix} n \\ r \end{smallmatrix} \right]$ be the number of ways to arrange n distinct objects around r indistinguishable circles such that each circle has at least one object. Show that

$$\left[\begin{smallmatrix} n \\ r \end{smallmatrix} \right] = \left[\begin{smallmatrix} n-1 \\ r-1 \end{smallmatrix} \right] + (n-1) \left[\begin{smallmatrix} n-1 \\ r \end{smallmatrix} \right].$$

These numbers are known as **Stirling Numbers of First kind**.

12. Given positive integers r and n with $r \leq n$, let $\left\{ \begin{smallmatrix} n \\ r \end{smallmatrix} \right\}$ be the number of ways to arrange n distinct objects in r indistinguishable box such that each box has at least one object. Show that

$$\left\{ \begin{smallmatrix} n \\ r \end{smallmatrix} \right\} = \left\{ \begin{smallmatrix} n-1 \\ r-1 \end{smallmatrix} \right\} + r \left\{ \begin{smallmatrix} n-1 \\ r \end{smallmatrix} \right\}.$$

These numbers are known as **Stirling Numbers of Second kind**.

13. Let $X = \{1, 2, \dots, n\}$. Show that the number of subsets of X having r elements, which contain no consecutive integers is $\binom{n-r+1}{r}$.

14. Find the number of ways of distributing r distinct objects into n identical boxes so that no box is empty and the objects in each box are arranged in a row.

15. Find the number of integer solutions to the equation $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 60$, in each of the following cases:

(i) $x_i > i - 1$ for each $i = 1, 2, \dots, 6$;

(ii) $x_1 \geq 2, x_2 \geq 5, 2 \leq x_3 \leq 7, x_4 \geq 1, x_5 \geq 3, x_6 \geq 2$.

16. Find the number of integer solutions to the equation: $x_1 + x_2 + x_3 + x_4 = 30$, in each of the following cases:

(i) $x_i > 0$ for each $i = 1, 2, 3, 4$;

(ii) $2 \leq x_1 \leq 7$ and $x_i \geq 0$ for each $i = 2, 3, 4$;

(iii) $x_1 \geq -5, x_2 \geq -1, x_3 > 1$ and $x_4 \geq 2$.

17. Find the number of quadruples (w, x, y, z) of non-negative integers which satisfy the inequality $w + x + y + z \leq 1992$.

18. Find the number of positive integer solutions to the equation: $(x_1 + x_2 + x_3)(y_1 + y_2 + y_3 + y_4) = 77$.

19. There are 5 ways to express 4 as a sum of 2 non-negative integers in which the order counts: $4 = 4 + 0 = 3 + 1 = 2 + 2 = 1 + 3 = 0 + 4$. What is the number of ways to express r as a sum of n non-negative integers in which the order counts?

20. There are 6 ways to express 5 as a sum of 3 positive integers in which the order counts: $5 = 3 + 1 + 1 = 2 + 2 + 1 = 2 + 1 + 2 = 1 + 3 + 1 = 1 + 2 + 2 = 1 + 1 + 3$. Given positive integers r and n with $r \geq n$, what is the number of ways to express r as a sum of n positive integers in which the order counts?

21. A positive integer d is said to be ascending if in its decimal representation: $d = d_m d_{m-1} \dots d_2 d_1$ we have $0 \leq d_m \leq d_{m-1} \leq \dots \leq d_2 \leq d_1$. For instance, 1337 and 2455566799 are ascending integers. Find the number of ascending integers which are less than 10^9 .

22. A positive integer d is said to be strictly ascending if in its decimal representation: $d = d_m d_{m-1} \dots d_2 d_1$ we have $0 < d_m < d_{m-1} < \dots < d_2 < d_1$. For instance, 145 and 23689 are strictly ascending integers. Find the number of strictly ascending integers which are less than 10^5 .

23. Prove that $\binom{n}{r} = \binom{n}{n-r}$, $r\binom{n}{r} = n\binom{n-1}{r-1}$, $\binom{n}{m}\binom{m}{r} = \binom{n}{r}\binom{n-r}{m-r}$.

24. Prove each of the following using binomial theorem and again using a combinatorial argument

(i) $\binom{n}{0} + \binom{n}{1} + \dots + \binom{n}{n-1} + \binom{n}{n} = 2^n$.

(ii) $\sum_{r \text{ odd}} \binom{n}{r} = \sum_{r \text{ even}} \binom{n}{r} = 2^{n-1}$.

(iii) $\sum_{r=0}^n r\binom{n}{r} = 1\binom{n}{1} + 2\binom{n}{2} + \dots + n\binom{n}{n} = n2^{n-1}$.

(iv) $\sum_{i=0}^r \binom{m}{i}\binom{n}{r-i} = \binom{m+n}{r}$.

25. Order the binomial coefficients $\binom{n}{r}$ where $0 \leq r \leq n$.

26. Let p be a prime number, then $p \mid \binom{p}{r}$ for $0 < r < p$.

27. Let $a_n = 6^n + 8^n$. Determine the remainder on dividing a_{83} by 49.

28. Prove that $9 \mid 4^n - 3n - 1$ for all positive integer n .

29. The number 4 can be expressed as a sum of one or more positive integers, taking order into account, in 8 ways: $4 = 1+3 = 3+1 = 2+2 = 1+1+2 = 1+2+1 = 2+1+1 = 1+1+1+1$. In how many ways can a positive integer n be so expressed?

30. Show that the number of distinct terms in the expansion of $(x_1 + \dots + x_m)^n$ is $\binom{n+m-1}{n}$.

31. Let p be a prime number. Then $\binom{n}{p} \equiv \lfloor \frac{n}{p} \rfloor \pmod{p}$.

32. $\binom{r}{r} + \binom{r+1}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1}$.

33. Show that for any set of 10 points chosen within a square whose sides are of length 3 units, there are two points in the set whose distance apart is at most $\sqrt{2}$.

34. Show that among any 5 points in an equilateral triangle of unit side length, there are 2 whose distance is at most $\frac{1}{2}$ units apart.

35. Given any set S of 9 points within a unit square, show that there always exist 3 distinct points in S such that the area of the triangle formed by these 3 points is less than or equal to $\frac{1}{8}$.

36. Show that given any set of 5 numbers, there are 3 numbers in the set whose sum is divisible by 3.

- 37.** Let A be a set of $n + 1$ elements. Show that there exist $a, b \in A$ with $a \neq b$ such that $n \mid (a - b)$.
- 38.** Show that among any group of n people, where $n \geq 2$, there are at least two people who know exactly the same number of people in the group (assuming that “knowing” is a symmetry relation).
- 39.** Prove that at a gathering of any six people some three of them are either mutual acquaintances or complete strangers to one another.
- 40.** Six points are in general position in space (no three in a line, no four in a plane). The fifteen line segments joining them in pairs are drawn, and then painted with some segments red and the rest blue. Prove that some triangle has all its sides the same colour.
- 41.** Seventeen people correspond by mail with one another - each one with all the rest. In their letters only three different topics are discussed. Each pair of correspondents deals with only one of these topics. Prove that there are at least three people who write to one another about the same topic.
- 42.** Let A be a set of m positive integers where $m > 1$. Show that there exists a non-empty subset B of A such that the sum of the elements of B is divisible by m .
- 43.** Let $A \subseteq \{1, 2, \dots, 2n\}$ such that $|A| = n + 1$. Show that there exist $a, b \in A$, with $a \neq b$ such that a, b are co-prime.
- 44.** Let a, b, c, d be integers. Show that $12 \mid (a - b)(a - c)(a - d)(b - c)(b - d)(c - d)$.
- 45.** From 52 positive integers we can select two such that their sum or difference is divisible by 100.
- 46.** Let $S = \{1, 2, \dots, 500\}$. Find number of integers in S which are
- (i) divisible by 2, 3 or 5.
 - (ii) none of 2, 3, 5.
 - (iii) exactly one of 2, 3, 5.
 - (iv) exactly two of 2, 3, 5.
 - (v) all of 2, 3, 5.